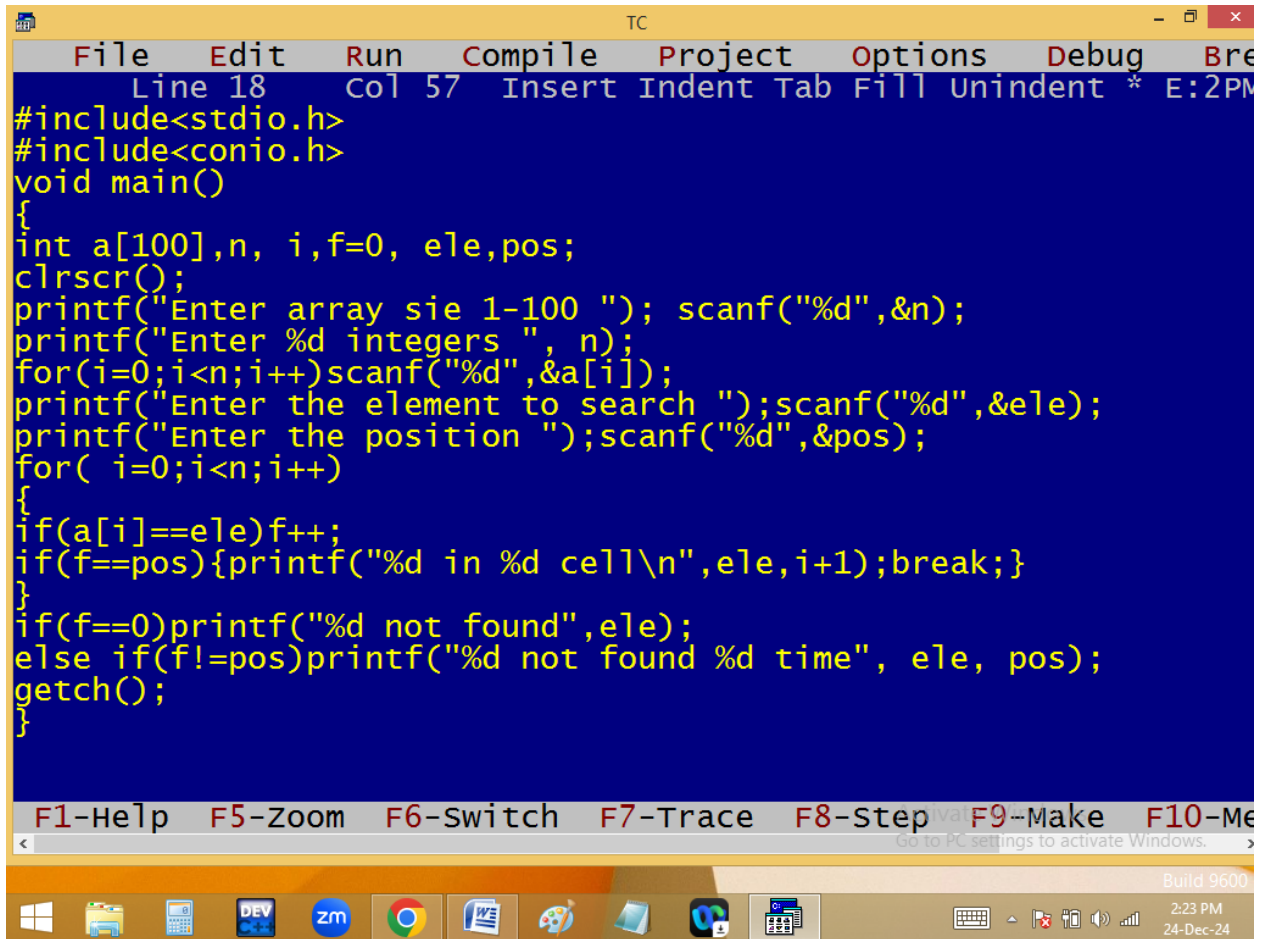


Finding nth occurrence of array element:



The screenshot shows the Turbo C++ (TC) IDE with a C program for finding the nth occurrence of an array element. The program is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a[100],n, i,f=0, ele,pos;
    clrscr();
    printf("Enter array size 1-100 "); scanf("%d",&n);
    printf("Enter %d integers ", n);
    for(i=0;i<n;i++)scanf("%d",&a[i]);
    printf("Enter the element to search ");scanf("%d",&ele);
    printf("Enter the position ");scanf("%d",&pos);
    for( i=0;i<n;i++)
    {
        if(a[i]==ele)f++;
        if(f==pos){printf("%d in %d cell\n",ele,i+1);break;}
    }
    if(f==0)printf("%d not found",ele);
    else if(f!=pos)printf("%d not found %d time", ele, pos);
    getch();
}
```

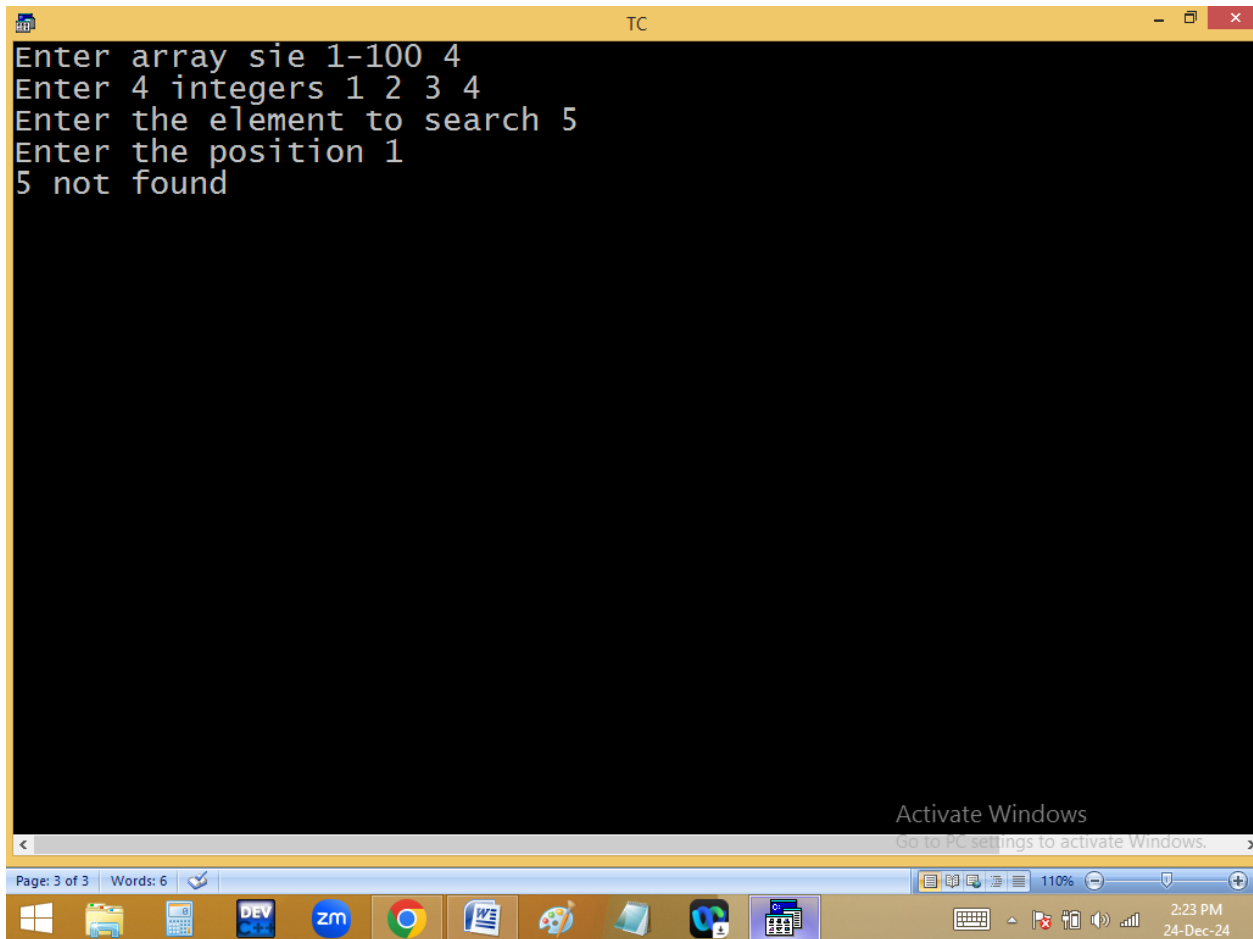
The IDE interface includes a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break), a status bar (Line 18, Col 57, Insert, Indent, Tab, Fill, Unindent, * E:2PM), and a toolbar with function keys (F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, F10-Me). The Windows taskbar at the bottom shows the Start button, several application icons (File Explorer, Calculator, DEV, zm, Chrome, Word, Paint, VS Code, Excel), and system tray icons (network, volume, date/time: 2:23 PM 24-Dec-24, Build 9600).

```
TC
Enter array sie 1-100 5
Enter 5 integers 1 2 3 4 5
Enter the element to search 1
Enter the position 2
1 not found 2 time
```

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Page: 1 of 1 Words: 6

2:23 PM
24-Dec-24

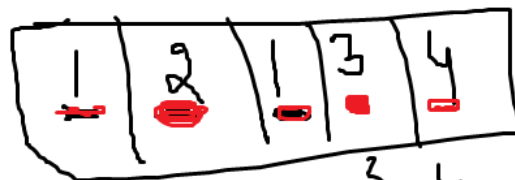


```
TC
Enter array size 1-100 5
Enter 5 integers 1 2 3 1 4
Enter the element to search 1
Enter the position 2
1 in 4 cell
```

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Page: 3 of 3 Words: 6 110%

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24-Dec-24



	0	1	2	3	4	pos	ele	f
	↑		+1			2	1	0
			3				2	1
							2	0
							2	1

if(f==0) p(not found);
else if(f!=pos) p(ele not found %d time",pos);

1!=2

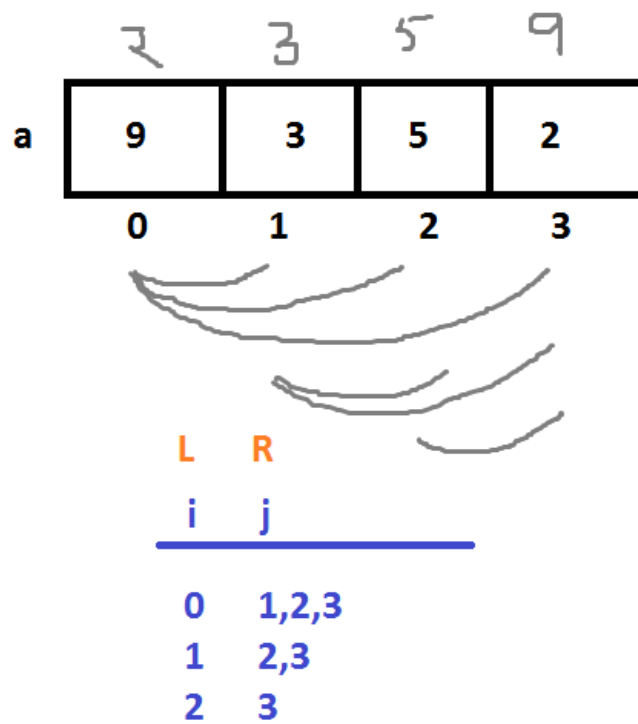
2

2

Sorting: Arranging data in order.

Selection sort in ascending order:

9	3	5	2
3	9	5	2
2	9	5	3
2	5	9	3
2	3	9	5
2	3	5	9



```
TC
File Edit Run Compile Project Options Debug Break
Line 17 Col 63 Insert Indent Tab Fill Unindent * E:2PM
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n, t,i,j;
clrscr();
printf("Enter array size 1-100 "); scanf("%d",&n);
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for( i=0;i<=n-2;i++)
{
for( j=i+1;j<=n-1;j++)
{
if(a[i]>a[j]){t=a[i]; a[i]=a[j]; a[j]=t;}
}
}
printf("Sorted elements ");for(i=0;i<n;i++)printf("%4d",a[i]);
getch();
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me

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2:47 PM 24-Dec-24

```

Enter array size 1-100 9
Enter 9 integers 4 0 -5 9 1 4 -2 7 6
Sorted elements -5 -2 0 1 4 4 6 7 9

```

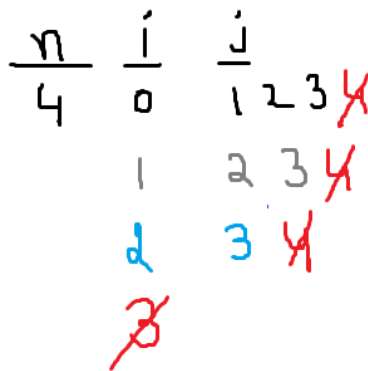
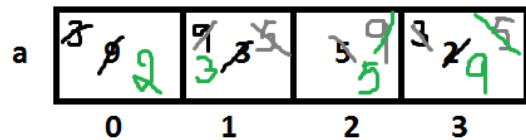
Activate Windows
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Page: 6 of 7 Words: 16 110% 2:47 PM 24-Dec-24

```

for( i=0; i<=n-2; i++ )
{
  for( j=i+1; j<=n-1; j++)
  { L R
    if(a[i]>a[j])
    {       
      t=a[i];
      a[i]=a[j];
      a[j]=t;
    }
  }
}

```



L	R
i	j
0	1,2,3
1	2,3
2	3

Descending order:

```
TC
File Edit Run Compile Project Options Debug Break
Line 14 Col 9 Insert Indent Tab Fill Unindent * E:2PM
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n, t,i,j;
clrscr();
printf("Enter array size 1-100 "); scanf("%d",&n);
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for( i=0;i<=n-2;i++)
{
for( j=i+1;j<=n-1;j++)
{
if(a[i]<a[j]){t=a[i]; a[i]=a[j]; a[j]=t;}
}
}
printf("sorted elements ");for(i=0;i<n;i++)printf("%4d",a[i]);
getch();
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me

Page: 8 of 8 Words: 18 110% 2:48 PM 24-Dec-24


```
TC
Enter array size 1-100 7
Enter 7 integers 4 0 2 9 -4 7 0
Sorted elements 9 7 4 2 0 0 -4
```

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Page: 8 of 8 Words: 18 110%

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24-Dec-24

Bubble sort:

L	R
i	j, j+1
0	0-1, 1-2, 2-3
1	0-1, 1-2
2	0-1



9	3	5	2
3	9	5	2
3	5	9	2
3	5	2	9
3	2	5	9
2	3	5	9

$$j \leftarrow \frac{n - i - 2}{4 - 0 - 2} = 2$$

$$4 - 1 - 2 = 1$$

$$4 - 2 - 2 = 0$$

L	R
i	j, j+1
0	0-1, 1-2, 2-3
1	0-1, 1-2
2	0-1

```
TC
File Edit Run Compile Project Options Debug Bre
Line 17 Col 46 Insert Indent Tab Fill Unindent * E:2PM
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n, t,i,j;
clrscr();
printf("Enter array size 1-100 "); scanf("%d",&n);
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for( i=0;i<=n-2;i++)
{
for( j=0;j<=n-i-2;j++)
{
if(a[j]>a[j+1]){t=a[j]; a[j]=a[j+1]; a[j+1]=t;}
}
}
printf("sorted elements ");for(i=0;i<n;i++)printf("%4d",a[i]);
getch();
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me

Go to PC settings to activate Windows.

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```
TC
Enter array size 1-100 9
Enter 9 integers 2 0 7 1 9 5 7 4 -2
Sorted elements -2 0 1 2 4 5 7 7 9
```

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Page: 11 of 11 Words: 20 110%

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24-Dec-24

Descending order:

```
TC
File Edit Run Compile Project Options Debug Bre
Line 14 Col 9 Insert Indent Tab Fill Unindent * E:2PM
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n, t,i,j;
clrscr();
printf("Enter array size 1-100 "); scanf("%d",&n);
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for( i=0;i<=n-2;i++)
{
for( j=0;j<=n-i-2;j++)
{
if(a[j]<a[j+1]){t=a[j]; a[j]=a[j+1]; a[j+1]=t;}
}
}
printf("sorted elements ");for(i=0;i<n;i++)printf("%4d",a[i]);
getch();
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me

Go to PC settings to activate Windows.

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```
TC
Enter array size 1-100 9
Enter 9 integers 3 0 -4 7 1 -8 4 2 9
Sorted elements 9 7 4 3 2 1 0 -4 -8
```

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Go to PC settings to activate Windows.

Page: 13 of 14 Words: 22 110% 3:19 PM 24-Dec-24

Arrange even elements in ascending order and odd elements in descending order:

```
TC
File Edit Run Compile Project Options Debug Bre
Line 14 Col 9 Insert Indent Tab Fill Unindent * E:2PM
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n, t,i,j;
clrscr();
printf("Enter array size 1-100 "); scanf("%d",&n);
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for( i=0;i<=n-2;i++)
{
for( j=0;j<=n-i-2;j++)
{
if(a[j]>a[j+1]){t=a[j]; a[j]=a[j+1]; a[j+1]=t;}
}
}
printf("Even elements ");
for(i=0;i<n;i++)if(a[i]%2==0)printf("%4d",a[i]);
printf("\nodd elements ");
for(i=n-1;i>=0;i--)if(a[i]%2!=0)printf("%4d",a[i]);
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me
Go to PC settings to activate Windows.
Page: 14 of 15 Words: 34 110%
3:25 PM
24-Dec-24
```

```
TC
Enter array size 1-100 10
Enter 10 integers 3 0 5 -5 9 3 -9 6 1 2
Even elements    0  2  6
Odd elements     9  5  3  3  1 -5 -9
```

Activate Windows
Go to PC settings to activate Windows.

Page: 15 of 16 Words: 34 110% 3:25 PM 24-Dec-24

Half elements ascending and half descending order:


```
TC
File Edit Run Compile Project Options Debug Break
Line 1 Col 19 Insert Indent Tab Fill Unindent * E:2PM
#include<stdio.h> #include<conio.h>
void main()
{
int a[100],n, t,i,j;
clrscr();
printf("Enter array size 1-100 "); scanf("%d",&n);
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for( i=0;i<=n-2;i++)
{
for( j=0;j<=n-i-2;j++)
{
if(a[j]>a[j+1]){t=a[j]; a[j]=a[j+1]; a[j+1]=t;}
}
}
printf("Ascending elements ");
for(i=0;i<n/2;i++)printf("%4d",a[i]);
printf("\nDescending elements ");
for(i=n-1;i>=n/2;i--)printf("%4d",a[i]);
getch();
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me

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```
TC
Enter array size 1-100 10
Enter 10 integers 3 0 12 7 8 -4 0 3 -8 1
Ascending elements -8 -4 0 0 1
Descending elements 12 8 7 3 3
```

Activate Windows
Go to PC settings to activate Windows.

Page: 17 of 17 Words: 41 110% 3:30 PM 24-Dec-24

Find the 2nd min, max elements of array:

```
TC
File Edit Run Compile Project Options Debug Bre
Line 19 Col 50 Insert Indent Tab Fill Unindent * E:2PM
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n, t,i,j;
clrscr();
printf("Enter array size 1-100 "); scanf("%d",&n);
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for( i=0;i<=n-2;i++)
{
for( j=0;j<=n-i-2;j++)
{
if(a[j]>a[j+1]){t=a[j]; a[j]=a[j+1]; a[j+1]=t;}
}
}
printf("elements ");
for(i=0;i<n;i++)printf("%4d",a[i]);
printf("\n2nd min=%d, 2nd max=%d", a[1], a[n-2]);_
getch();
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me

Go to PC settings to activate Windows.

Page: 18 of 18 Words: 49 110%

3:35 PM 24-Dec-24

```
TC
Enter array size 1-100 7
Enter 7 integers 2 0 1 -4 8 -2 6
elements -4 -2 0 1 2 6 8
2nd min=-2, 2nd max=6
```

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Page: 19 of 19 Words: 49 110%

3:35 PM
24-Dec-24

```
TC
File Edit Run Compile Project Options Debug Bre
Line 9 Col 49 Insert Indent Tab Fill Unindent * E:2PM
#include<stdio.h> #include<conio.h>
void main()
{
int a[100],n, t,i,j;clrscr();
printf("Enter array size 1-100 "); scanf("%d",&n);
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for( i=0;i<=n-2;i++)
{
for( j=0;j<=n-i-2;j++)
{if(a[j]>a[j+1]){t=a[j]; a[j]=a[j+1]; a[j+1]=t;}}
}
printf("elements ");
for(i=0;i<n;i++)printf("%4d",a[i]);
for(i=1;i<n;i++)
if(a[i]>a[0]){printf("\n2nd min %d\n",a[i]);break;}

for(i=n-2;i>=0;i--)
if(a[i]<a[n-1]){printf("2nd max %d",a[i]);break;}
getch();
}

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me
Go to PC settings to activate Windows.
16 x 36px 1894 x 763px 100%
Windows taskbar with icons for File Explorer, Calculator, DEV, zm, Chrome, Word, Paint, and others. System tray shows 3:48 PM on 24-Dec-24.
```

```
TC
Enter array size 1-100 7
Enter 7 integers 1 0 3 9 0 2 5 9 6
elements      0    0    1    2    3    5    9
2nd min 1
2nd max 5
```

Activate Windows
Go to PC settings to activate Windows.

Page: 22 of 22 | Words: 49 | 110%

3:48 PM
24-Dec-24

```
TC
Enter array size 1-100 7
Enter 7 integers 1 1 2 2 3 4 4
elements      1      1      2      2      3      4      4
2nd min 2
2nd max 3
```

Activate Windows
Go to PC settings to activate Windows.

Page: 23 of 23 | Words: 49 | 110%

3:49 PM
24-Dec-24

```
TC
Enter array size 1-100 9
Enter 9 integers 3 0 1 2 1 7 5 7 6
elements      0    1    1    2    3    5    6    7    7
2nd min 1
2nd max 6
```

Activate Windows
Go to PC settings to activate Windows.

Page: 20 of 21 Words: 49 110%

3:48 PM
24-Dec-24


```
TC
Enter array size 1-100 7
Enter 7 integers 1 1 2 2 3 4 4
elements      1      1      2      2      3      4      4
2nd min 2
2nd max 3
```

Activate Windows
Go to PC settings to activate Windows.

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3:53 PM
24-Dec-24